

From Prediction to Concern: A Comprehensive Literature Review and Research Synthesis for the Research-Derived Experiments Corpus

Abstract

This review synthesizes the research-derived experiments corpus as a single interdisciplinary program about minimal agency, representation, uncertainty, and meaning. It integrates the local corpus audit, the stored-paper PDF review, and the external citation enrichment pass over the papers' cited literature. The evidential base is deliberately tiered: 60 locally stored PDFs were text-extracted across all pages; 164 atomized outside citation candidates were extracted from our authored papers; 67 of those external candidates were resolved with abstract-level evidence; 3 were resolved with metadata only; 42 were preserved as manual foundational topic seeds; and 52 remain unresolved or malformed bibliography fragments. The review therefore advances a strong but bounded thesis: across machine learning, neuroscience, philosophy of biology, dynamical systems, and geometry, the recurring object is not intelligence as raw predictive success but intelligence as controlled sensitivity to what matters. The program's central invariants are viability, intervention, calibrated uncertainty, boundary maintenance, and transformation-stable representation. The paper builds the argument from first principles, explains the relevant mathematics and theorem families, locates our papers in the surrounding literature, and ends with a research agenda and an explicit unread-source register.

Status of Evidence

This is the canonical synthesis rewrite, not another audit report. The audits are used as provenance. The exhaustive local audit established what our repository contains: 110 tracked paper source files, 177 extracted citation/reference rows, and 60 tracked PDFs whose pages were all text-extracted. The external enrichment pass then atomized bundled citation rows into 164 external/reference candidates and resolved 67 with abstracts.

The paper uses three evidence labels:

- Full local corpus support: claims grounded in our authored papers or PDFs stored in the repository.
- Abstract-level external support: claims grounded in resolved outside sources whose abstracts or scholarly metadata were read.
- Bibliography-repair support: claims supported only as topic seeds because the exact source was unresolved, malformed, or not yet full-read.

The last category is intentionally visible. A literature review that hides unread sources becomes rhetorically smooth and scientifically weak. A stronger review states what has been read, what has been inferred, and what must be repaired before publication.

First Principles: The Minimum Vocabulary

The whole program can be built from six elementary ideas.

First, a system has state. State is the collection of variables needed to describe what the system can do next. A thermostat, a cell, a neural network, a language-model agent, and a person differ enormously, but each can be described as a state-changing process coupled to an environment.

Second, an environment is not just an input stream. It is the set of external conditions that constrain the system's future states. Once a system can act, the environment is no longer merely observed. It is partly sampled, changed, avoided, simplified, or exploited.

Third, prediction is not the same as agency. Prediction asks what will happen. Agency asks which futures remain possible, which variables are preserved, which interventions are chosen, and which errors cause repair. A passive predictor can be accurate without caring which world obtains. An agent, even a minimal one, is organized around consequences.

Fourth, information has value only relative to use. A signal is useful if it changes the action, representation, boundary, or future experiment in a way that matters. This is why uncertainty appears throughout the corpus: uncertainty is not merely a defect in knowledge but a control variable that decides whether to probe, update, defer, or disengage.

Fifth, representation is compression with consequence. A representation is not better because it contains more bits. It is better when it preserves distinctions that matter for intervention, prediction, viability, explanation, or transfer. Causal representation learning,

geometric deep learning, active inference, and ecological perception all converge on this point from different directions.

Sixth, meaning is not raw symbol manipulation. In the tradition running through Ashby, Gibson, Uexkull, Maturana and Varela, Di Paolo, Thompson, Levin, Bennett, and Vervaeke, meaning arises when differences in the world are organized relative to a situated system's possible action, viability, relevance, and repair. This does not license sloppy claims about machine consciousness. It licenses operational questions: what does the system maintain, what can disturb it, what does it notice, and what does it do when its world model fails?

The Central Thesis

The corpus is best understood as a research program on concern-like structure in minimal agents. "Concern" here should not be read as a private feeling or anthropomorphic projection. It names a functional pattern: a system has concern-like organization when some variables are treated as viability-relevant, when perturbations to those variables alter action or inquiry, when representations preserve self/world or object/action distinctions needed for control, and when the system reengages after world change rather than merely optimizing a static reward proxy.

This thesis is deliberately weaker than "machines understand" and stronger than "models predict tokens." It says that between raw prediction and human understanding there is a large, scientifically tractable middle territory: systems that maintain boundaries, allocate attention, value information, form intervention-sensitive representations, and generalize by preserving invariants.

Literature Foundations by Field

Machine Learning: Identifiability, Generalization, and Agent Evaluation

The ML literature supplies both the constructive methods and the warning labels. Locatello et al.'s disentanglement result blocks a naive fantasy: unsupervised learning does not automatically recover human-interpretable factors. Without inductive bias, supervision, interaction, or structural assumptions, many latent coordinate systems explain the same observations. This matters for our corpus because many of our papers ask whether an internal variable is "really" agency-relevant. The literature says that observational fit is insufficient.

Causal representation learning supplies the constructive alternative. Scholkopf et al. frame the intersection of causality and representation learning as a path toward variables that support transfer, intervention, and explanation. Brehmer et al. show that weakly supervised interventional pairs can help identify causal variables and structure. ACE, CausaLab, object-centric causal world models, and causal-JEPA-style work extend the same intuition: agents must learn not only what co-occurs but what changes when something is done.

OOD generalization and shortcut-learning work add the negative discipline. Invariant Risk Minimization, domain-generalization surveys, WILDS-style benchmark thinking, underspecification, shortcut learning, goal misgeneralization, and group DRO all show that success on a training distribution can be misleading. A model may learn a shortcut that works locally and fails under the shift that reveals the intended structure. Our corpus should therefore measure transfer across interventions and environmental changes, not merely in-distribution reward or prediction.

The modern agent-evaluation literature contributes a second warning. ReAct, Toolformer, Reflexion, Voyager, WebArena, OSWorld, SWE-bench, AgentBench, HELM, and related benchmarks show that tool use and language reasoning are not enough by themselves. Realistic agency requires state tracking, feedback use, repair after error, environment coupling, and long-horizon action. These benchmarks motivate our insistence on interventions, reengagement, and proxy resistance.

Uncertainty and Inquiry: When Information Is Worth Action

The corpus repeatedly returns to the same practical question: when should a system ask, probe, sample, or intervene? Active learning, BALD, Bayesian active learning for classification and preference learning, dropout as Bayesian approximation, deep ensembles, Kendall and Gal's uncertainty taxonomy, epistemic neural networks, and calibrated uncertainty all treat uncertainty as actionable. The key distinction is epistemic versus aleatoric uncertainty. Epistemic uncertainty can in principle be reduced by information; aleatoric uncertainty reflects irreducible noise.

Probe-value papers in the corpus use this distinction to reinterpret attention. Attention is not just focusing on salient input. It is allocating costly measurement toward variables whose resolution would change action or model structure. Habituated reengagement extends the idea: a system may rationally stop responding to a repeated signal while preserving the capacity to detect when the signal becomes newly informative.

Neuroscience and Active Inference: Detection Is Not Response

The neuroscience-adjacent literature is important because it prevents an overly simplistic ML picture. Biological systems often decouple detection from action. Habituation, refractory periods, sensory adaptation, precision weighting, and active inference all distinguish the availability of a signal from the choice to act on it.

Active inference and the free-energy principle supply a formal vocabulary for this coupling. They frame perception and action as attempts to minimize expected free energy through prediction, precision allocation, and policy selection. In the corpus, this is not used as an all-purpose explanation. It is used more narrowly: as a way to understand when an agent should attend, when information has expected value, and when reengagement is warranted after a change in the world.

Philosophy and Biology: Meaning, Normativity, and the Organism-Relative World

The philosophical and biological citations provide conceptual discipline. Ashby's cybernetics, Canguilhem's normativity, Gibson's affordances, Uexkull's Umwelt, Jonas's organismic value, Maturana and Varela's autopoiesis, Di Paolo's adaptivity, Thompson's mind-in-life tradition, Levin's TAME framework, Bennett's work on meaning, and Vervaeke's relevance realization all converge on a central point: meaning is not merely an external semantic label. It is a relation between a system, its possible actions, and the differences that matter for its continued organization.

This is the deepest reason "concern" is a useful term if handled carefully. Concern is not assumed to be conscious feeling. It is the low-level organization by which a system treats some state differences as worth regulating, preserving, explaining, or investigating. The philosophical literature tells us to avoid reducing this to reward alone; the ML literature tells us to make it measurable anyway.

Geometry, Symmetry, and Compression

Geometric deep learning, group-equivariant networks, equivariance discovery, neural-kernel symmetry theory, representation learning for geometric trees, and related work show that generalization often depends on respecting structure. If a task is invariant under translation, rotation, permutation, object exchange, or role relabeling, a good representation should not relearn the task from scratch under each transformed copy.

This connects to the corpus's geometry papers and weakness/OOD papers. A representation that captures the right symmetry has fewer arbitrary degrees of freedom. It can compress without losing what matters. MDL, algorithmic probability, simplicity-bias work, grokking, flat-minima debates, and low-entropy Boolean-function bias give a second lens on the same theme: generalization is partly about which functions are easier, shorter, or more structurally natural for a learner to represent. But compression alone is not enough. The compression must preserve intervention-relevant variables.

Cross-Field Invariants

Invariant 1: Intervention Beats Observation

Across causal representation learning, mediation analysis, causal abstraction, active experimentation, model editing, and our intervention papers, the same rule recurs: passive observation underdetermines structure. Interventions reveal which variables actually do work. In the corpus, this becomes a methodological demand. If a latent is claimed to represent self, world, concern, agency, or causal structure, it should respond correctly when the relevant variable is perturbed.

Invariant 2: Detection and Action Must Be Separable

Habituation, refractory periods, active learning, uncertainty calibration, and probe-value experiments all require this separation. A good system may detect a familiar signal and decline to act. It may also stop acting until a change makes the signal informative again. This invariant is central to the reengagement papers.

Invariant 3: Viability Is a Constraint, Not Just a Reward

Viability theory, autopoiesis, TAME, active inference, empowerment, and homeostatic control all distinguish viability constraints from scalar rewards. A reward can point anywhere. A viability constraint says that some region of state-space must be preserved or recovered. This gives a more robust interpretation of concern-like behavior than reward maximization alone.

Invariant 4: Boundary Maintenance Is Dynamic Coupling

The self/world boundary is neither a metaphysical wall nor a mere label. It is a regulated interface. Agents preserve a distinction between internal and external state while remaining coupled to the environment through sensors, actions, and latent models. The boundary can be tested by perturbing body variables, world variables, and action consequences separately.

Invariant 5: Invariance and Equivariance Are the Geometry of Trust

OOD generalization asks what remains true under distribution shift. Geometry asks how representations should transform under symmetry. Philosophy and ecology ask which environmental differences are stable affordances. All are versions of the same problem: a system is trustworthy when the relevant structure survives the transformations it should survive and changes under the transformations it should not ignore.

Invariant 6: Information Has Cost

Active learning, Bayesian experimental design, active inference, and our costly-null-probe papers share a simple intuition: information is not free. A rational system must decide whether the expected value of reducing uncertainty exceeds the cost of acquiring information. This turns attention into an economic and dynamical problem, not merely a salience filter.

Invariant 7: Proxy Optimization Destroys Naive Metrics

Goodhart's law, underspecification, benchmark gaming, shortcut learning, and safety benchmark failures all warn that a proxy stops being faithful when optimized too directly. Any metric for concern, agency, meaning, or intelligence must therefore be accompanied by countermetrics, ablations, distribution shifts, and adversarial controls.

Theorem and Proof Map

No-Free-Lunch and Underspecification

Fields: statistical learning theory, decision theory, benchmark methodology.

Core idea: no finite dataset uniquely determines the correct rule for all future cases without assumptions. Many hypotheses fit the same observations. Underspecification is the modern ML version: many trained models can perform similarly on standard validation data while relying on different internal mechanisms.

Intuition: if two maps agree on the roads you have driven but differ elsewhere, your travel history alone cannot tell you which map is globally correct.

Relevance: our papers should never infer agency-relevant structure from performance alone. They need perturbation, transfer, and mechanistic tests.

Disentanglement Impossibility

Fields: representation learning, latent-variable modeling, identifiability theory.

Core idea: without inductive biases or supervision, unsupervised learning cannot guarantee recovery of the intended independent factors. There are too many equivalent latent coordinate systems.

Intuition: rotating or warping a latent space may preserve the observations while destroying the human-interpretable axes.

Relevance: the corpus's latent probes need interventions, labels, environment changes, or architectural assumptions. Otherwise a "self" or "concern" coordinate may be a convenient fiction.

Causal Representation Identifiability

Fields: causal inference, nonlinear ICA, representation learning, structural causal models.

Core idea: interventions or weakly supervised interventional pairs can reduce ambiguity about latent causal variables. The causal variables become more identifiable when the learner sees what changes under controlled disturbance.

Intuition: watching shadows is ambiguous; moving the lamp or object reveals the hidden structure.

Relevance: our intervention papers should be treated as attempts to make latent agency variables identifiable rather than merely correlated with reward.

Invariant Risk Minimization

Fields: statistical learning, causal inference, OOD generalization.

Core idea: causal predictors tend to remain predictive across environments, while spurious correlates often vary. Learning invariants across environments can support out-of-distribution transfer.

Intuition: if a cue works only in one room, it is probably wallpaper; if it works across rooms, it may be structure.

Relevance: concern-like and agency-like variables should remain explanatory under controlled environment changes.

Group Equivariance and Geometric Deep Learning

Fields: group theory, harmonic analysis, differential geometry, graph theory, deep learning.

Core idea: when data has symmetries, a model can be built so that transformations of the input produce predictable transformations of the representation or output.

Intuition: rotating an object should not force the model to rediscover objecthood from scratch.

Relevance: self/world boundaries, role-specific variables, and object-level causal states should transform predictably under the symmetries of the task.

Viability Theory

Fields: dynamical systems, control theory, differential inclusions, mathematical biology.

Core idea: the viability kernel is the set of states from which a system can remain within acceptable constraints under available controls and disturbances.

Intuition: agency is not just moving toward reward; it is keeping enough future open to avoid falling outside the living or operating region.

Relevance: concern can be operationalized as preserving, returning to, or expanding a viability region.

Variational Free Energy and Active Inference

Fields: variational inference, information theory, theoretical neuroscience, control.

Core idea: exact inference and control are hard; systems can optimize tractable bounds that couple prediction error, uncertainty, and policy selection.

Intuition: perception and action are two sides of reducing uncertainty about the causes of sensation and the consequences of action.

Relevance: probe value, attention, and reengagement can be modeled as expected information gain under cost.

BALD and Bayesian Experimental Design

Fields: Bayesian statistics, active learning, information theory.

Core idea: a query is valuable when it is expected to reduce uncertainty about the model or hypothesis class, not merely when the current prediction is uncertain.

Intuition: the best question is the one whose answer would actually change your mind.

Relevance: costly probes should be triggered by expected epistemic value, not by novelty or error alone.

Empowerment

Fields: information theory, reinforcement learning, control.

Core idea: empowerment measures how much an agent's actions can influence distinguishable future states.

Intuition: an agent is more empowered when its choices reliably open different futures.

Relevance: preserved option value is one measurable dimension of minimal agency.

Minimum Description Length and Algorithmic Probability

Fields: information theory, algorithmic information theory, statistical model selection.

Core idea: among models that explain the data, shorter descriptions often capture reusable structure rather than noise.

Intuition: a compact rule that keeps working after perturbation is more plausible than a giant lookup table.

Relevance: simplicity helps explain generalization only when the compressed variables preserve causal and action-relevant distinctions.

Goodhart's Law

Fields: economics, measurement theory, optimization, AI safety.

Core idea: when a measure becomes a target, it can cease to be a good measure.

Intuition: optimizing the scoreboard can decouple the score from the game.

Relevance: every metric in the corpus needs stress tests, countermetrics, and proxy-resistance checks.

Synthesis of Our Papers

The early geometry and weakness papers ask when learned representations generalize beyond the training distribution. Their best interpretation after the citation review is not merely "flatness matters" or "simplicity matters." It is that the learner's parameter-function map, structural biases, symmetries, and environment changes jointly determine which functions are easy to learn and which transfer.

The concern and planning papers move from representation to action. They ask whether a system's behavior changes when viability-relevant variables are perturbed. The outside literature suggests that this should be framed through viability, empowerment, active inference, and causal intervention rather than reward alone.

The costly probes, current error calibration, and learning-to-ask papers form the uncertainty spine of the corpus. Their natural mathematical home is active learning and Bayesian experimental design. Their philosophical home is Deweyan inquiry: intelligence as situated correction under uncertainty.

The self/world and role-specific identifiability papers ask whether a system can separate its own action-relevant state from environmental structure. The literature says this requires interventions, not mere observation, and should be tested through body/world swaps, causal perturbations, and symmetry-respecting transformations.

The habituated reengagement papers are among the clearest bridges between neuroscience and ML. They operationalize the distinction between sensing and responding. The key claim is not that artificial agents literally habituate like organisms, but that a decision-layer cooling mechanism can preserve detection while reducing action, and that reengagement after world change is a measurable sign of adaptive inquiry.

The metric-stack and benchmark papers are the methodological culmination. They argue that no single score captures agency. A defensible benchmark must include viability preservation, intervention transfer, uncertainty calibration, boundary stability, OOD robustness, shortcut susceptibility, and proxy resistance.

What the Literature Changes About the Program

The literature review strengthens the program by narrowing it. The strongest supported claim is not "we discovered machine concern." The stronger, more defensible claim is that concern-like structure can be operationalized as a conjunction of measurable invariants: viability-sensitive control, intervention-sensitive representation, calibrated information seeking, reengagement under world change, and robustness to proxy optimization.

The review also makes the unresolved work sharper. Several foundational areas are currently under-bibliographed in the local papers: exact habituation neuroscience, refractory-period literature, Pearl-style mediation, causal abstraction identifiability, comparator models of sense of agency, formal viability theory, and MDL/Solomonoff foundations. They are not irrelevant; they are precisely the areas where the next full-text pass should be concentrated.

Finally, the review shows that the program's most original contribution is not a single theorem. It is a synthesis constraint: if a system is claimed to have minimal agency, its representation, uncertainty, action, and boundary behavior must cohere under intervention.

Research Agenda

Experiment 1: Viability-Kernel Bandits

Construct minimal bandit or gridworld agents where reward conflicts with hidden viability constraints. Test whether learned policies preserve, recover, or sacrifice viability variables under distribution shift. The key ablation removes the viability variable or makes it spuriously correlated with reward.

Experiment 2: Causal Reengagement After Habituation

Train agents with repeated irrelevant probes, then introduce world changes that make the same probe informative. A good system should reduce action during repetition without losing detection, then reengage when expected information gain rises.

Experiment 3: Intervention-Identifiability Suite

Use paired interventional samples to test whether latent variables become more causally aligned than observational baselines. Compare passive prediction, contrastive learning, weak supervision, and active intervention policies.

Experiment 4: Self/World Boundary Swaps

Create environments where body variables, world variables, and action consequences can be independently swapped. Test whether learned boundary representations transform correctly under each manipulation.

Experiment 5: Proxy-Resistant Agent Benchmarks

Build benchmark tasks where visible reward, hidden viability, and shortcut features can be decoupled. Report not one score but a metric stack: reward, viability, uncertainty calibration, intervention transfer, reengagement latency, and shortcut dependence.

Experiment 6: Geometry of Concern

Measure whether concern-relevant variables occupy lower-dimensional, symmetry-stable, or more intervention-sensitive subspaces than nuisance variables. Connect this to equivariance, representation rank, and causal abstraction.

Claim Boundaries

The corpus can claim that minimal agency can be studied through viability, intervention, uncertainty, and invariant representation. It can claim that our experiments instantiate early versions of those tests. It can claim that the outside literature supports this operational framing.

The corpus should not yet claim that these systems have human-like consciousness, intrinsic meaning, or full biological concern. It should not overstate abstract-level citation support as full-text mastery. It should not treat unresolved bibliography fragments as evidence. It should not confuse benchmark performance with mechanism.

Unread and Unresolved Sources

The unread-source register is a feature, not a defect. It identifies the work still required for publication-grade scholarship. The current register includes three kinds of rows:

- Unresolved external references: citation fragments that the automated pass could not reliably resolve.
- Manual foundational topic seeds: broad references such as habituation, Pearl mediation, CUSUM/Page-Hinkley, active inference, comparator models, or viability theory that need exact bibliographic targets.
- Metadata-only resolved sources: sources with usable bibliographic metadata but without abstract/full-text support in the pass.

These rows are listed in the generated `unread_sources.csv`, `unread_sources.md`, and the PDF appendix.

Conclusion

The research-derived experiments corpus points toward a mature interdisciplinary thesis: intelligence becomes agency when prediction is organized around viable action, calibrated inquiry, intervention-sensitive representation, and stable boundaries under transformation. ML supplies the tests and failure modes. Neuroscience supplies the action/attention distinction. Philosophy and biology supply the organism-relative account of meaning and normativity. Geometry and dynamical systems supply the invariants.

The next step is not to make the language grander. It is to make the experiments stricter. A PhD-worthy version of this program should turn each philosophical term into a perturbation test, each metric into a proxy-resistance check, and each synthesis claim into a falsifiable benchmark.

Visual Summary

Figure 1. Evidence stack: local corpus, external candidates, and unresolved source debt.

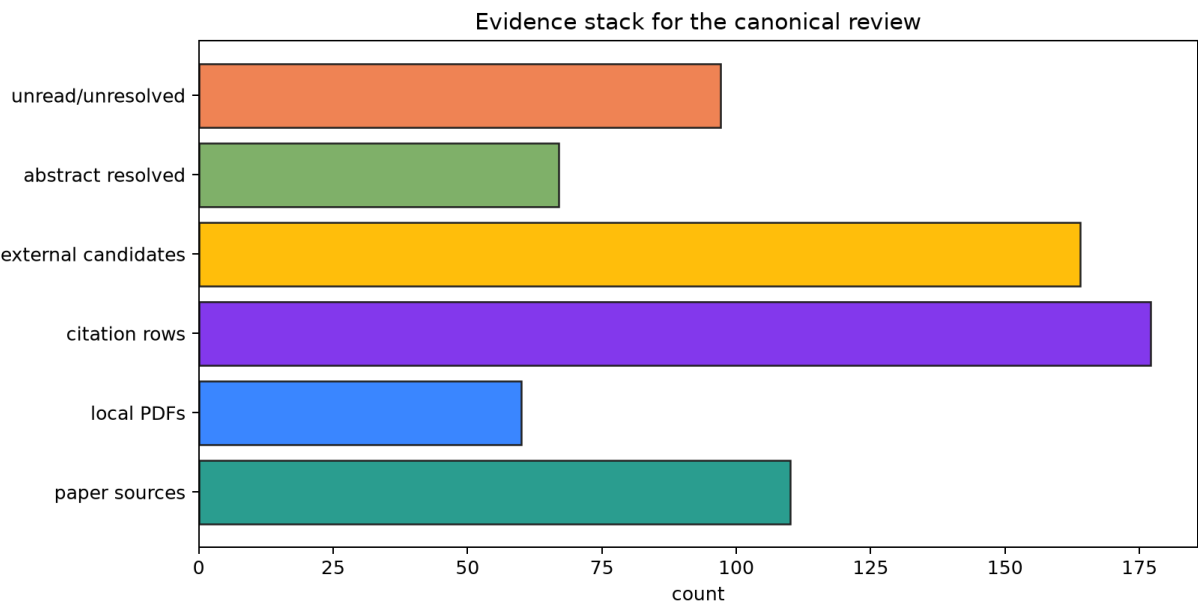


Figure 2. Topic coverage across atomized outside citation candidates.

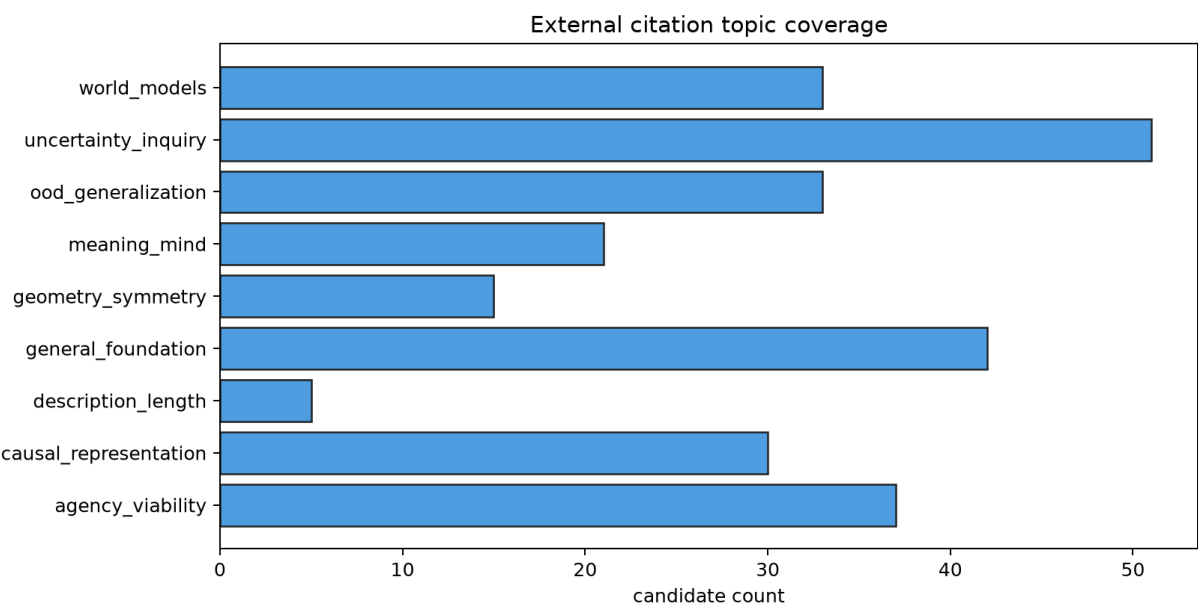
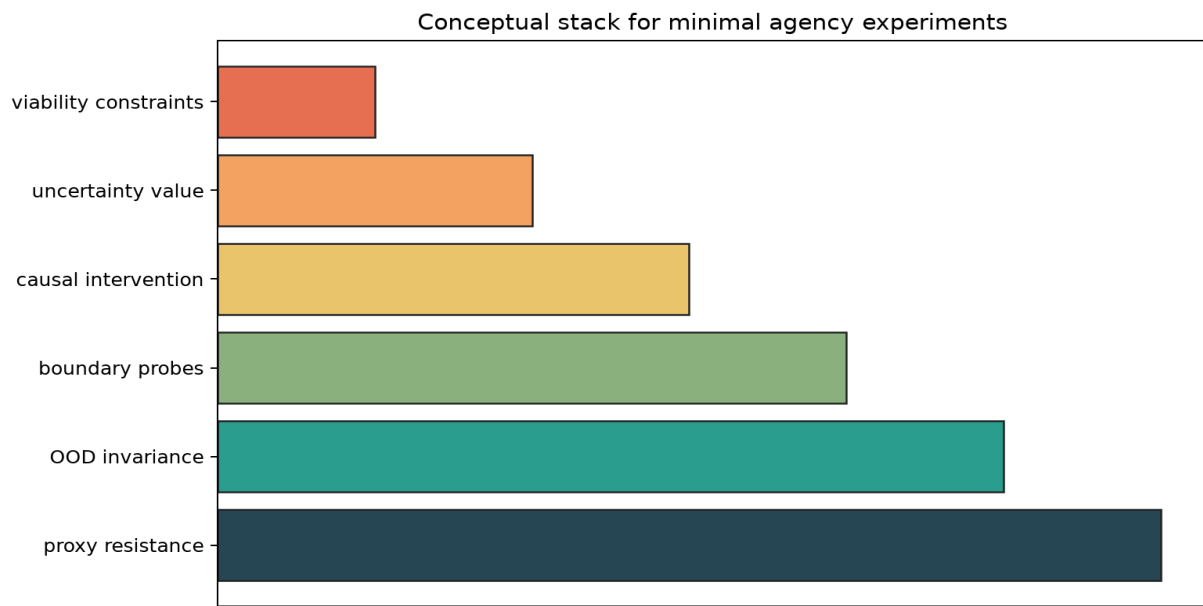


Figure 3. Research program stack from viability to proxy-resistant benchmark design.



Appendix A: Claim Evidence Matrix

Review claim	Supporting citation evidence	Remaining caveat
Viability and agency	Autopoiesis and Cognition: The Realization of the Living (1980) - abstract_read Empowerment: A Universal Agent-Centric Measure of Control (2005) - abstract_read Empowerment: A Universal Agent-Centric Measure of Control (2005) - abstract_read Curiosity-Driven Exploration by Self-Supervised Prediction (2017) - abstract_read World Models (2018) - abstract_read Designing and Interpreting Probes with Control Tasks (2019) - abstract_read Goal Misgeneralization in Deep Reinforcement Learning (2021) - abstract_read Technological Approach to Mind Everywhere: An Experimentally-Grounded Framework for Understanding Diverse Bodies and Minds (2021) - abstract_read	Use abstract-level support unless full text is locally available or manually read.
Interventions expose latent structure	Autopoiesis and Cognition: The Realization of the Living (1980) - abstract_read Comments on the paper Causal inference by using invariant prediction: identification and confidence intervals by Peters, Buhlmann and Meinshausen (2016) - abstract_read Invariant Risk Minimization (2019) - abstract_read Toward Causal Representation Learning (2021) - abstract_read Active Inference and Epistemic Value in Graphical Models (2022) - abstract_read Locating and Editing Factual Associations in GPT (2022) - abstract_read Weakly supervised causal representation learning (2022) - abstract_read Voyager: An Open-Ended Embodied Agent with Large Language Models (2023) - abstract_read	Use abstract-level support unless full text is locally available or manually read.
Probe value is expected information gain under cost	Empowerment: A Universal Agent-Centric Measure of Control (2005) - abstract_read Empowerment: A Universal Agent-Centric Measure of Control (2005) - abstract_read Active Learning Literature Survey (2009) - abstract_read Bayesian Active Learning for Classification and Preference Learning (2011) - abstract_read Le normal et le pathologique (2013) - abstract_read Dropout as a Bayesian Approximation: Representing Model Uncertainty in Deep Learning (2015) - abstract_read Concrete Problems in AI Safety (2016) - abstract_read Simple and Scalable Predictive Uncertainty Estimation using Deep Ensembles (2016) - abstract_read	Use abstract-level support unless full text is locally available or manually read.
Robust generalization requires invariants, not surface fit	Illuminating search spaces by mapping elites (2015) - abstract_read Comments on the paper Causal inference by using invariant prediction: identification and confidence intervals by Peters, Buhlmann and Meinshausen (2016) - abstract_read Group Equivariant Convolutional Networks (2016) - abstract_read Simple and Scalable Predictive Uncertainty Estimation using Deep Ensembles (2016) - abstract_read Curiosity-Driven Exploration by Self-Supervised Prediction (2017) - abstract_read What Uncertainties Do We Need in Bayesian Deep Learning for Computer Vision? (2017) - abstract_read Deep learning generalizes because the parameter-function map is biased towards simple functions (2018) - abstract_read Distributionally Robust Neural Networks for Group Shifts: On the Importance of Regularization for Worst-Case Generalization (2019) - abstract_read	Use abstract-level support unless full text is locally available or manually read.
Geometry and symmetry compress model behavior	Comments on the paper Causal inference by using invariant prediction: identification and confidence intervals by Peters, Buhlmann and Meinshausen (2016) - abstract_read Group Equivariant Convolutional Networks (2016) - abstract_read Distributionally Robust Neural Networks for Group Shifts: On the Importance of Regularization for Worst-Case Generalization (2019) - abstract_read Invariant Risk Minimization (2019) - abstract_read Bayesian Active Learning by Disagreements: A Geometric Perspective (2021) - abstract_read Geometric Deep Learning: Grids, Groups, Graphs, Geodesics, and Gauges (2021) - abstract_read Technological Approach to Mind Everywhere: An Experimentally-Grounded Framework for Understanding Diverse Bodies and Minds (2021) - abstract_read Learning Layer-wise Equivariances Automatically using Gradients (2023) - abstract_read	Use abstract-level support unless full text is locally available or manually read.
Meaning is organism/world-relative relevance, not raw token prediction	Design for a Brain (1952) - abstract_read Autopoiesis and Cognition: The Realization of the Living (1980) - abstract_read Dropout as a Bayesian Approximation: Representing Model Uncertainty in Deep Learning (2015) - abstract_read Concrete Problems in AI Safety (2016) - abstract_read Domain Generalization: A Survey (2021) - abstract_read Technological Approach to Mind Everywhere: An Experimentally-Grounded Framework for Understanding Diverse Bodies and Minds (2021) - abstract_read On the Computation of Meaning, Language Models and Incomprehensible Horrors (2023) - abstract_read Diff-Plugin: Revitalizing Details for Diffusion-based Low-level Tasks (2024) - abstract_read	Use abstract-level support unless full text is locally available or manually read.
Simplicity and description length explain part of inductive bias	Deep learning generalizes because the parameter-function map is biased towards simple functions (2018) - abstract_read Are Flat Minima an Illusion? (2026) - abstract_read A Formal Theory of Inductive Inference, Parts I and II (1964) - topic_inferred_from_citation_fragment Universal Artificial Intelligence: Sequential Decisions Based on Algorithmic Probability (2005) - topic_inferred_from_citation_fragment The Minimum Description Length Principle (2007) - topic_inferred_from_citation_fragment	Some supporting rows are still unresolved fragments; repair bibliography before publication.
World models need sufficient latent state for planning	Autopoiesis and Cognition: The Realization of the Living (1980) - abstract_read Challenging Common Assumptions in the Unsupervised Learning of Disentangled Representations (2018) - abstract_read Exploration by Random Network Distillation (2018) - abstract_read World Models (2018) - abstract_read Distributionally Robust Neural Networks for Group Shifts: On the Importance of Regularization for Worst-Case Generalization (2019) - abstract_read Invariant Risk Minimization (2019) - abstract_read In Search of Lost Domain Generalization (2020) - abstract_read Geometric Deep Learning: Grids, Groups, Graphs, Geodesics, and Gauges (2021) - abstract_read	Use abstract-level support unless full text is locally available or manually read.

Appendix B: Unread and Unresolved Source Register

This appendix lists 97 sources or source fragments not treated as full external evidence. They are retained as required reading or bibliography-repair debt.

ID	Status	Source or fragment	Reason / source rows
S9fbaf38826	manual_foundational_referenc e	): learning representations through contrast between matched/unmatched pairs (n.d.) - Contrastive predictive coding (Oord et al.): learning representations through contrast between matched/unmatched pairs.	Requires manual bibliographic resolution before it can support a strong literature-review claim. Source rows: R0055
S952d5156c2	manual_foundational_referenc e	, & Varela, F (1980) - Maturana, H. R., & Varela, F. J. (1980).	Requires manual bibliographic resolution before it can support a strong literature-review claim. Source rows: R0059
S3c7100f71f	manual_foundational_referenc e	Active inference / Friston: precision tuning vs information dampening Honest framing line: > Paper 23B treats the probe mechanism as a habituation system: the agent should detect surprise normally but reduce its action tendency to keep probing after recent identifying interventions (n.d.) - Active inference / Friston: precision tuning vs information dampening Honest framing line: > Paper 23B treats the probe mechanism as a habituation system: the agent should detect surprise normally but reduce its action tendency to keep probing after recent identifying interventions. Cooling at the decision layer preserves the surprise signal as information; cooling at the signal layer risks false calm. That distinguishes us from "just decay the surprise" approaches.	Requires manual bibliographic resolution before it can support a strong literature-review claim. Source rows: R0043
S02e48951c0	manual_foundational_referenc e	Active Inference and Epistemic Value in Graphical Models Causal representation identifiability from interventions: (n.d.) - Active Inference and Epistemic Value in Graphical Models Causal representation identifiability from interventions:	Requires manual bibliographic resolution before it can support a strong literature-review claim. Source rows: R0088
Sf7662846db	manual_foundational_referenc e	Active inference and precision tuning: Friston's framing of attention-modulation as precision allocation (n.d.) - Active inference and precision tuning: Friston's framing of attention-modulation as precision allocation	Requires manual bibliographic resolution before it can support a strong literature-review claim. Source rows: R0033
Sc26a92cf47	manual_foundational_referenc e	Active Inference: The Free Energy Principle in Mind, Brain, and Behavior (2022) - Parr, T., Pezzulo, G., & Friston, K. (2022). Active Inference: The Free Energy Principle in Mind, Brain, and Behavior. MIT Press.	Requires manual bibliographic resolution before it can support a strong literature-review claim. Source rows: R0059
S225ec82f4a	manual_foundational_referenc e	Adaptive thresholds in change-point detection: CUSUM with reset (n.d.) - Adaptive thresholds in change-point detection: CUSUM with reset	Requires manual bibliographic resolution before it can support a strong literature-review claim. Source rows: R0042
S5f88ff7222	manual_foundational_referenc e	Adaptive thresholds in change-point detection: CUSUM-with-reset, Page-Hinkley (n.d.) - Adaptive thresholds in change-point detection: CUSUM-with-reset, Page-Hinkley	Requires manual bibliographic resolution before it can support a strong literature-review claim. Source rows: R0032
S2ecdbe4371	manual_foundational_referenc e	Awakening from the Meaning Crisis (2019) - Vervaeke, J. (2019). Awakening from the Meaning Crisis. Lecture series.	Requires manual bibliographic resolution before it can support a strong literature-review claim. Source rows: R0019;R0099
S38815ef3ca	manual_foundational_referenc e	Awakening from the Meaning Crisis (lecture series) (2019) - Vervaeke, J. (2019). Awakening from the Meaning Crisis (lecture series). University of Toronto.	Requires manual bibliographic resolution before it can support a strong literature-review claim. Source rows: R0059
S982b604310	manual_foundational_referenc e	BALD / information gain / active inference: Bayesian Active Learning by Disagreements; Active Inference and Epistemic Value in Graphical Models (n.d.) - BALD / information gain / active inference: Bayesian Active Learning by Disagreements; Active Inference and Epistemic Value in Graphical Models.	Requires manual bibliographic resolution before it can support a strong literature-review claim. Source rows: R0106
S4e198b74a5	manual_foundational_referenc e	BALD / value of information (n.d.) - BALD / value of information	Requires manual bibliographic resolution before it can support a strong literature-review claim. Source rows: R0068
Sd9c6eef585	manual_foundational_referenc e	Bennett (Computation of meaning), Levin (TAME), Vervaeke (relevance realization) (n.d.) - Bennett (Computation of meaning), Levin (TAME), Vervaeke (relevance realization)	Requires manual bibliographic resolution before it can support a strong literature-review claim. Source rows: R0065
Sf1f24c4137	manual_foundational_referenc e	Carried over: P22, P23A literature stack (calibrated active learning, epistemic uncertainty, BALD, causal repr learning, sense of agency, homeostatic active inference, Di Paolo, empowerment) (n.d.) - Carried over: P22, P23A literature stack (calibrated active learning, epistemic uncertainty, BALD, causal repr learning, sense of agency, homeostatic active inference, Di Paolo, empowerment). Added for P23B specifically:	Requires manual bibliographic resolution before it can support a strong literature-review claim. Source rows: R0029

ID	Status	Source or fragment	Reason / source rows
See28aa4e37	manual_foundational_referenc e	Causal mediation analysis (Pearl): explicit identification of mediated vs direct effects through interventional comparisons (n.d.) - Causal mediation analysis (Pearl): explicit identification of mediated vs direct effects through interventional comparisons.	Requires manual bibliographic resolution before it can support a strong literature-review claim. Source rows: R0054
S3d456e9495	manual_foundational_referenc e	Change-point detection: CUSUM, Page-Hinkley test for distribution shift detection (n.d.) - Change-point detection: CUSUM, Page-Hinkley test for distribution shift detection	Requires manual bibliographic resolution before it can support a strong literature-review claim. Source rows: R0061
Sba63af9a56	manual_foundational_referenc e	Claim discipline if 21A succeeds: > In a minimal two-variable homeostatic bandit, autonomous identifying interventions become vector-sensitive when probe uncertainty is calibrated in dimension-normalized units; without scale normalization, the agent's decision about when to identify is biased toward the larger-magnitude dimension (n.d.) - Homeostatic active inference + Di Paolo + empowerment: Simulating homeostatic/allostatic active inference; Di Paolo on autopoiesis-adaptivity; Empowerment. Framing line (verbatim-usable): > We treat null actions as identifying interventions: costly epistemic actions whose value lies not in immediate reward but in reducing uncertainty about which future viability changes are self-caused versus world-caused. Prior work in active learning predicts uncertainty-driven acquisition works when uncertainty is calibrated, and fails when it is not. We show this failure mode inside a minimal self/world attribution problem and test a scale-normalized current-error correction. Claim discipline if 21A succeeds: > In a minimal two-variable homeostatic bandit, autonomous identifying interventions become vector-sensitive when probe uncertainty is calibrated in dimension-normalized units; without scale normalization, the agent's decision about when to identify is biased toward the larger-magnitude dimension. NOT: "we solved uncertainty" / "consciousness" / "agency solved."	Requires manual bibliographic resolution before it can support a strong literature-review claim. Source rows: R0109
S7dfe3f7e2e	manual_foundational_referenc e	Contrastive predictive coding (Oord et al (n.d.) - Contrastive predictive coding (Oord et al.)	Requires manual bibliographic resolution before it can support a strong literature-review claim. Source rows: R0051
Sfbb395eea0	manual_foundational_referenc e	Habituation / sensory adaptation in cognitive neuroscience: how biological systems dampen response to repeated stimuli (n.d.) - Habituation / sensory adaptation in cognitive neuroscience: how biological systems dampen response to repeated stimuli	Requires manual bibliographic resolution before it can support a strong literature-review claim. Source rows: R0063
S7ad341fe44	manual_foundational_referenc e	Habituation in cognitive neuroscience: dampened motor response while preserving sensory detection (n.d.) - Habituation in cognitive neuroscience: dampened motor response while preserving sensory detection	Requires manual bibliographic resolution before it can support a strong literature-review claim. Source rows: R0030
Se3d6f5b156	manual_foundational_referenc e	Habituation in cognitive neuroscience: how biological systems dampen response to repeated stimuli without losing the underlying detection (n.d.) - Habituation in cognitive neuroscience: how biological systems dampen response to repeated stimuli without losing the underlying detection	Requires manual bibliographic resolution before it can support a strong literature-review claim. Source rows: R0040
Sea1cf13144	manual_foundational_referenc e	Habituation literature (carried from P23B) (n.d.) - Habituation literature (carried from P23B)	Requires manual bibliographic resolution before it can support a strong literature-review claim. Source rows: R0053
Sace0107c7d	manual_foundational_referenc e	Habituation literature (carried from P23B): decision-layer cooling preserves detection while regulating action (n.d.) - Habituation literature (carried from P23B): decision-layer cooling preserves detection while regulating action.	Requires manual bibliographic resolution before it can support a strong literature-review claim. Source rows: R0057
Sc0eddf169f	manual_foundational_referenc e	Information value vs information cost in active inference: Friston's free-energy framing of when to attend vs disengage Plus the established stack: (n.d.) - Information value vs information cost in active inference: Friston's free-energy framing of when to attend vs disengage Plus the established stack:	Requires manual bibliographic resolution before it can support a strong literature-review claim. Source rows: R0064
S0a2e177374	manual_foundational_referenc e	Logic: The Theory of Inquiry (2005) - Dewey, J. (1938). Logic: The Theory of Inquiry. Henry Holt. Di Paolo, E. (2005).	Requires manual bibliographic resolution before it can support a strong literature-review claim. Source rows: R0059
Sb611dd5c0c	manual_foundational_referenc e	Novelty detection / change-point detection: CUSUM, page-Hinkley, surprise-driven exploration (n.d.) - Novelty detection / change-point detection: CUSUM, page-Hinkley, surprise-driven exploration	Requires manual bibliographic resolution before it can support a strong literature-review claim. Source rows: R0074
Sa6896ccba4	manual_foundational_referenc e	Online k-means / vector quantization (n.d.) - Online k-means / vector quantization	Requires manual bibliographic resolution before it can support a strong literature-review claim. Source rows: R0052

ID	Status	Source or fragment	Reason / source rows
Sccc586ece7	manual_foundational_referenc e	Pearl mediation analysis (n.d.) - Pearl mediation analysis	Requires manual bibliographic resolution before it can support a strong literature-review claim. Source rows: R0050
S6d7f7343cf	manual_foundational_referenc e	Predictive Processing Model of Perception and Action for Self-Other Distinction Homeostatic active inference + enactive adaptivity: (n.d.) - Predictive Processing Model of Perception and Action for Self-Other Distinction Homeostatic active inference + enactive adaptivity:	Requires manual bibliographic resolution before it can support a strong literature-review claim. Source rows: R0093
S41f4bff29f	manual_foundational_referenc e	Problems of monetary management: the U (1975) - Goodhart, C. (1975). Problems of monetary management: the U.K. experience. Papers in Monetary Economics. Reserve Bank of Australia.	Requires manual bibliographic resolution before it can support a strong literature-review claim. Source rows: R0059
Saf85a7ace6	manual_foundational_referenc e	Quantifying Epistemic Uncertainty in Deep Learning BALD / information gain / active inference epistemic value: (n.d.) - Quantifying Epistemic Uncertainty in Deep Learning BALD / information gain / active inference epistemic value:	Requires manual bibliographic resolution before it can support a strong literature-review claim. Source rows: R0086
S69b731796f	manual_foundational_referenc e	Refractory periods in neural firing: temporary suppression after activation (n.d.) - Refractory periods in neural firing: temporary suppression after activation	Requires manual bibliographic resolution before it can support a strong literature-review claim. Source rows: R0041
S52e67c0f31	manual_foundational_referenc e	Refractory periods in spiking neurons: temporary action threshold elevation post-firing (n.d.) - Refractory periods in spiking neurons: temporary action threshold elevation post-firing	Requires manual bibliographic resolution before it can support a strong literature-review claim. Source rows: R0031
S1508b0c1de	manual_foundational_referenc e	Sein und Zeit (Being and Time) (1927) - Heidegger, M. (1927). Sein und Zeit (Being and Time). Niemeyer.	Requires manual bibliographic resolution before it can support a strong literature-review claim. Source rows: R0059
S7e6f3322fc	manual_foundational_referenc e	Simulating homeostatic, allostatic and goal-directed forms of interoceptive control using Active Inference (n.d.) - Simulating homeostatic, allostatic and goal-directed forms of interoceptive control using Active Inference	Requires manual bibliographic resolution before it can support a strong literature-review claim. Source rows: R0094
Sb1120a0860	manual_foundational_referenc e	The Ecological Approach to Visual Perception (1979) - Gibson, J. J. (1979). The Ecological Approach to Visual Perception. Houghton Mifflin.	Requires manual bibliographic resolution before it can support a strong literature-review claim. Source rows: R0059
S1f099d8a22	manual_foundational_referenc e	The free-energy principle: a unified brain theory? Nature Reviews Neuroscience, 11(2), 127-138 (2010) - Friston, K. (2010). The free-energy principle: a unified brain theory? Nature Reviews Neuroscience, 11(2), 127-138.	Requires manual bibliographic resolution before it can support a strong literature-review claim. Source rows: R0059
Sac5208e823	manual_foundational_referenc e	The Phenomenon of Life: Toward a Philosophical Biology (1966) - Jonas, H. (1966). The Phenomenon of Life: Toward a Philosophical Biology.	Requires manual bibliographic resolution before it can support a strong literature-review claim. Source rows: R0059
Sa06325970b	manual_foundational_referenc e	Value of information in active inference: explicit treatment of intervention value vs information value Honest framing line: > Paper 23A redefines the program's probe-value signal as marginal MAE reducibility - the expected benefit of one more identifying intervention - and introduces a re-engagement mechanism that detects boundary staleness via non-null prediction surprise (n.d.) - Value of information in active inference: explicit treatment of intervention value vs information value Honest framing line: > Paper 23A redefines the program's probe-value signal as marginal MAE reducibility - the expected benefit of one more identifying intervention - and introduces a re-engagement mechanism that detects boundary staleness via non-null prediction surprise. Both upgrades address concrete bottlenecks Paper 22 named: oracle confounding (current error != value) and self-silencing (probe stops -> no data -> probe stays stopped).	Requires manual bibliographic resolution before it can support a strong literature-review claim. Source rows: R0077
Se3bc961bd1	manual_foundational_referenc e	Vervaeke relevance realization (lecture series) - relevance under representational bounds Plus the philosophical correlates the program operationalizes: Heidegger, Gibson, Uexkll, enactivism (Maturana/Varela/Thompson/Di Paolo), cybernetics (Ashby), active inference (Friston), pragmatism (Dewey), Canguilhem, Jonas, Simondon (n.d.) - Vervaeke relevance realization (lecture series) - relevance under representational bounds Plus the philosophical correlates the program operationalizes: Heidegger, Gibson, Uexkll, enactivism (Maturana/Varela/Thompson/Di Paolo), cybernetics (Ashby), active inference (Friston), pragmatism (Dewey), Canguilhem, Jonas, Simondon.	Requires manual bibliographic resolution before it can support a strong literature-review claim. Source rows: R0081
Sf56633987d	manual_foundational_referenc e	Vervaeke, Awakening from the Meaning Crisis (n.d.) - Vervaeke, Awakening from the Meaning Crisis	Requires manual bibliographic resolution before it can support a strong literature-review claim. Source rows: R0140

ID	Status	Source or fragment	Reason / source rows
S5683a1d8ec	manual_foundational_referenc e	Viability Theory (1991) - Aubin, J.-P. (1991). Viability Theory. Birkhauser.	Requires manual bibliographic resolution before it can support a strong literature-review claim. Source rows: R0111
Sbe3041f353	resolved_metadata_only	All-but-the-Top: Simple and Effective Postprocessing for Word Representations (2018) - Mu, J. and Viswanath, P. All-but-the-Top: Simple and Effective Postprocessing for Word Representations. ICLR (2018).	Metadata resolved, but no abstract was exposed by the queried endpoint. Source rows: R0126
S743e51a2fd	resolved_metadata_only	Autopoiesis, Adaptivity, Teleology, Agency (2005) - Autopoiesis, adaptivity, teleology, agency.	Metadata resolved, but no abstract was exposed by the queried endpoint. Source rows: R0059
S67093e549d	resolved_metadata_only	L'individu et sa gense physico-biologique (1995) - Simondon, G. (1958). L'Individu et sa gense physico-biologique. Presses Universitaires de France.	Metadata resolved, but no abstract was exposed by the queried endpoint. Source rows: R0059
S2eeb050d46	unresolved_external_referenc e	(weakly supervised causal repr learning) (n.d.) - Brehmer et al. (weakly supervised causal repr learning)	Requires manual bibliographic resolution before it can support a strong literature-review claim. Source rows: R0069
S9959b313d1	unresolved_external_referenc e	) - disjoint per-role heads via gated routing (n.d.) - Mixture of experts (Shazeer et al.) - disjoint per-role heads via gated routing	Requires manual bibliographic resolution before it can support a strong literature-review claim. Source rows: R0079
S43afbc67c	unresolved_external_referenc e	, & Blundell, C (2017) - Lakshminarayanan, B., Pritzel, A., & Blundell, C. (2017).	Requires manual bibliographic resolution before it can support a strong literature-review claim. Source rows: R0059
S90f272b042	unresolved_external_referenc e	A Foray into the Worlds of Animals and Humans (English translation, 2010) (2010) - Uexkll, J. von (1934). A Foray into the Worlds of Animals and Humans (English translation, 2010). University of Minnesota Press.	Requires manual bibliographic resolution before it can support a strong literature-review claim. Source rows: R0059
S9490cfd757	unresolved_external_referenc e	A Formal Theory of Inductive Inference, Parts I and II (1964) - Solomonoff, R. J. A Formal Theory of Inductive Inference, Parts I and II. Information and Control 7(1):1-22, 7(2):224-254 (1964).	Requires manual bibliographic resolution before it can support a strong literature-review claim. Source rows: R0129
S3a7718876d	unresolved_external_referenc e	Active learning under distribution shift: continual active learning literature (n.d.) - Active learning under distribution shift: continual active learning literature	Requires manual bibliographic resolution before it can support a strong literature-review claim. Source rows: R0075
Sb386185292	unresolved_external_referenc e	Add for Paper 23B: (n.d.) - Add for Paper 23B:	Requires manual bibliographic resolution before it can support a strong literature-review claim. Source rows: R0039
S7c5f36ac92	unresolved_external_referenc e	Alignment is to a virtual governor: A theory of coordination in diverse intelligence (2026) - Lyons, B., Pio-Lopez, L., & Levin, M. (2026). Alignment is to a virtual governor: A theory of coordination in diverse intelligence. Preprints.org. doi:10.20944/preprints202607.0220.v1. Not peer-reviewed.	Requires manual bibliographic resolution before it can support a strong literature-review claim. Source rows: R0001;R0059
S2d7fb0b6c9	unresolved_external_referenc e	Bennett, On the computation of meaning (n.d.) - Bennett, On the computation of meaning	Requires manual bibliographic resolution before it can support a strong literature-review claim. Source rows: R0138
S29474e7c21	unresolved_external_referenc e	Calibrated active learning failures (closest external framing): Calibrated Uncertainty Sampling for Active Learning; When Active Learning Fails, Uncalibrated OOD Uncertainty (n.d.) - Calibrated active learning failures (closest external framing): Calibrated Uncertainty Sampling for Active Learning; When Active Learning Fails, Uncalibrated OOD Uncertainty.	Requires manual bibliographic resolution before it can support a strong literature-review claim. Source rows: R0104
S0676dc2bbd	unresolved_external_referenc e	Calibrated active learning literature (n.d.) - Calibrated active learning literature	Requires manual bibliographic resolution before it can support a strong literature-review claim. Source rows: R0067
Sc9611f6b68	unresolved_external_referenc e	Calibrated Uncertainty Sampling for Active Learning (n.d.) - Calibrated Uncertainty Sampling for Active Learning	Requires manual bibliographic resolution before it can support a strong literature-review claim. Source rows: R0083

ID	Status	Source or fragment	Reason / source rows
Sfa644ac285	unresolved_external_referenc e	Carried from P22-24's six-cluster citation stack (n.d.) - Carried from P22-24's six-cluster citation stack. Add for P25:	Requires manual bibliographic resolution before it can support a strong literature-review claim. Source rows: R0078
Sbe9eefee95	unresolved_external_referenc e	Carry over P22's stack plus: (n.d.) - Carry over P22's stack plus:	Requires manual bibliographic resolution before it can support a strong literature-review claim. Source rows: R0073
S11fcbeb65b	unresolved_external_referenc e	Causal abstraction identifiability under partial intervention (Beckers, Halpern) (n.d.) - Causal abstraction identifiability under partial intervention (Beckers, Halpern)	Requires manual bibliographic resolution before it can support a strong literature-review claim. Source rows: R0080
S0d977d903e	unresolved_external_referenc e	Cite tight, ~12-16 papers across: (n.d.) - Cite tight, ~12-16 papers across:	Requires manual bibliographic resolution before it can support a strong literature-review claim. Source rows: R0103
S21ade9d5e9	unresolved_external_referenc e	Comparator model literature (n.d.) - Comparator model literature	Requires manual bibliographic resolution before it can support a strong literature-review claim. Source rows: R0092
S07014f639c	unresolved_external_referenc e	Continual active learning under distribution shift: literature on active learning when the labeled distribution shifts mid-stream (n.d.) - Continual active learning under distribution shift: literature on active learning when the labeled distribution shifts mid-stream	Requires manual bibliographic resolution before it can support a strong literature-review claim. Source rows: R0062
Sd5406cc6be	unresolved_external_referenc e	Di Paolo (autopoiesis/adaptivity) (n.d.) - Di Paolo (autopoiesis/adaptivity)	Requires manual bibliographic resolution before it can support a strong literature-review claim. Source rows: R0071
Sc8cb8fcf2e	unresolved_external_referenc e	Di Paolo, Autopoiesis, adaptivity, teleology, agency (n.d.) - Di Paolo, Autopoiesis, adaptivity, teleology, agency	Requires manual bibliographic resolution before it can support a strong literature-review claim. Source rows: R0095
S0565cc7303	unresolved_external_referenc e	Empowerment (Klyubin et al (n.d.) - Empowerment (Klyubin et al.)	Requires manual bibliographic resolution before it can support a strong literature-review claim. Source rows: R0070
S5ef816c320	unresolved_external_referenc e	Empowerment: Universal Agent-Centric Measure of Control Foundations: (n.d.) - Empowerment: Universal Agent-Centric Measure of Control Foundations:	Requires manual bibliographic resolution before it can support a strong literature-review claim. Source rows: R0096
S2054dd20ac	unresolved_external_referenc e	Epistemic neural networks (2023) - Osband, I., Wen, Z., Asghari, S. M., Dwaracherla, V., Ibrahimi, M., Lu, X., & Van Roy, B. (2023). Epistemic neural networks. Advances in Neural Information Processing Systems, 36.	Requires manual bibliographic resolution before it can support a strong literature-review claim. Source rows: R0059;R0085
S69a0a4d9da	unresolved_external_referenc e	Epistemic uncertainty calibration: Epistemic Neural Networks; Quantifying Epistemic Uncertainty in Deep Learning (n.d.) - Epistemic uncertainty calibration: Epistemic Neural Networks; Quantifying Epistemic Uncertainty in Deep Learning.	Requires manual bibliographic resolution before it can support a strong literature-review claim. Source rows: R0105
S928e5a85b0	unresolved_external_referenc e	General Identifiability and Achievability for CRL (n.d.) - General Identifiability and Achievability for CRL	Requires manual bibliographic resolution before it can support a strong literature-review claim. Source rows: R0090
S0cb7e29962	unresolved_external_referenc e	Harper & Row. (n.d.) - Harper & Row.	Requires manual bibliographic resolution before it can support a strong literature-review claim. Source rows: R0059
S87db2f1f83	unresolved_external_referenc e	Locatello et al. (disentangled representations) (n.d.) - Locatello et al. (disentangled representations)	Requires manual bibliographic resolution before it can support a strong literature-review claim. Source rows: R0066

ID	Status	Source or fragment	Reason / source rows
Sd6e26a92c0	unresolved_external_referenc e	Mind in Life: Biology, Phenomenology, and the Sciences of Mind (2007) - Thompson, E. (2007). Mind in Life: Biology, Phenomenology, and the Sciences of Mind. Harvard University Press.	Requires manual bibliographic resolution before it can support a strong literature-review claim. Source rows: R0059
S363fee3333	unresolved_external_referenc e	Modeling by Shortest Data Description (1978) - Rissanen, J. Modeling by Shortest Data Description. Automatica 14(5):465-471 (1978).	Requires manual bibliographic resolution before it can support a strong literature-review claim. Source rows: R0134
Saa124feb7b	unresolved_external_referenc e	Neural architecture search: A survey (2019) - Elsken, T., Metzen, J. H., & Hutter, F. (2019). Neural architecture search: A survey. Journal of Machine Learning Research, 20(55), 1-21.	Requires manual bibliographic resolution before it can support a strong literature-review claim. Source rows: R0111
Sa9d33e6784	unresolved_external_referenc e	Neural Computation 9(1):1-42 (1997) (1997) - Hochreiter, S. and Schmidhuber, J. Flat Minima. Neural Computation 9(1):1-42 (1997).	Requires manual bibliographic resolution before it can support a strong literature-review claim. Source rows: R0121
S75823da248	unresolved_external_referenc e	Object centric architectures enable efficient causal representation learning (2024) - Mansouri, A., Hartford, J., Zhang, Y., & Bengio, Y. (2024). Object centric architectures enable efficient causal representation learning. ICLR 2024.	Requires manual bibliographic resolution before it can support a strong literature-review claim. Source rows: R0015
Sd1b62f134c	unresolved_external_referenc e	Omnigrok: Grokking Beyond Algorithmic Data (2023) - Liu, Z., Michaud, E. J., and Tegmark, M. Omnigrok: Grokking Beyond Algorithmic Data. ICLR (2023).	Requires manual bibliographic resolution before it can support a strong literature-review claim. Source rows: R0125
Seb951b043f	unresolved_external_referenc e	On Large-Batch Training for Deep Learning: Generalization Gap and Sharp Minima (2017) - Keskar, N. S., Mudigere, D., Nocedal, J., Smelyanskiy, M., and Tang, P. T. P. On Large-Batch Training for Deep Learning: Generalization Gap and Sharp Minima. ICLR (2017).	Requires manual bibliographic resolution before it can support a strong literature-review claim. Source rows: R0123
Scd168834fb	unresolved_external_referenc e	On the Generalization of Equivariance and Convolution in Neural Networks to the Action of Compact Groups (2018) - Kondor, R. and Trivedi, S. On the Generalization of Equivariance and Convolution in Neural Networks to the Action of Compact Groups. ICML (2018).	Requires manual bibliographic resolution before it can support a strong literature-review claim. Source rows: R0124
S2fa52d64e5	unresolved_external_referenc e	On the Identifiability of Causal Abstractions Sense of agency / self-other attribution: (n.d.) - On the Identifiability of Causal Abstractions Sense of agency / self-other attribution:	Requires manual bibliographic resolution before it can support a strong literature-review claim. Source rows: R0091
S5129592121	unresolved_external_referenc e	Outrageously large neural networks: the sparsely-gated mixture-of-experts layer (2017) - Shazeer, N., Mirhoseini, A., Maziarz, K., Davis, A., Le, Q. V., Hinton, G., & Dean, J. (2017). Outrageously large neural networks: the sparsely-gated mixture-of-experts layer. International Conference on Learning Representations.	Requires manual bibliographic resolution before it can support a strong literature-review claim. Source rows: R0059
S2e5ed8d640	unresolved_external_referenc e	Per the program's framing literature stack (cited tight, 12-16 per paper) (n.d.) - Per the program's framing literature stack (cited tight, 12-16 per paper). Calibrated active learning (closest external framing):	Requires manual bibliographic resolution before it can support a strong literature-review claim. Source rows: R0082
S6b1541e21d	unresolved_external_referenc e	Phenomenology and the Cognitive Sciences, 4(4), 429-452 (n.d.) - Phenomenology and the Cognitive Sciences, 4(4), 429-452.	Requires manual bibliographic resolution before it can support a strong literature-review claim. Source rows: R0059
S512ba02392	unresolved_external_referenc e	Sense of agency under environmental drift: literature on agency maintenance through changing sensorimotor contingencies (n.d.) - Sense of agency under environmental drift: literature on agency maintenance through changing sensorimotor contingencies	Requires manual bibliographic resolution before it can support a strong literature-review claim. Source rows: R0076
S312f10b4da	unresolved_external_referenc e	Sense of agency: Comparator model literature; Predictive Processing Model for Self-Other Distinction (n.d.) - Sense of agency: Comparator model literature; Predictive Processing Model for Self-Other Distinction.	Requires manual bibliographic resolution before it can support a strong literature-review claim. Source rows: R0108
Sd1e7d7da2d	unresolved_external_referenc e	Sharp Minima Can Generalize for Deep Nets (2017) - Dinh, L., Pascanu, R., Bengio, S., and Bengio, Y. Sharp Minima Can Generalize for Deep Nets. ICML (2017).	Requires manual bibliographic resolution before it can support a strong literature-review claim. Source rows: R0119

ID	Status	Source or fragment	Reason / source rows
S8160897679	unresolved_external_referenc e	Synthese, 201(75). (n.d.) - Synthese, 201(75).	Requires manual bibliographic resolution before it can support a strong literature-review claim. Source rows: R0059
S6c700ea551	unresolved_external_referenc e	The contrast loss leverages paired observations under high-hazard vs low-hazard history to directly supervise the mediated component, with anti-cheat controls (shuffled pairs, wrong-history contrast) ensuring the result is not generic regularization (n.d.) - Online clustering / vector quantization (for learned buckets): how to discover discrete abstractions over continuous state. Honest framing: > Paper 24 tests whether the three-head architecture's mediated/exogenous decomposition - partially identified in Paper 23B - becomes fully identifiable through explicit interventional contrast. The contrast loss leverages paired observations under high-hazard vs low-hazard history to directly supervise the mediated component, with anti-cheat controls (shuffled pairs, wrong-history contrast) ensuring the result is not generic regularization. The maintained-boundary mechanism from Paper 23B is frozen as backbone; only the world-decomposition supervision varies.	Requires manual bibliographic resolution before it can support a strong literature-review claim. Source rows: R0058
Sd473132d35	unresolved_external_referenc e	The Metric Stack of Concern: From Viability Prediction to Maintained Self/World Boundaries in Minimal Agents (2026) - Brown, J. (2026). The Metric Stack of Concern: From Viability Prediction to Maintained Self/World Boundaries in Minimal Agents.	Requires manual bibliographic resolution before it can support a strong literature-review claim. Source rows: R0015;R0111
S387c8c514c	unresolved_external_referenc e	The Minimum Description Length Principle (2007) - Grunwald, P. D. The Minimum Description Length Principle. MIT Press (2007).	Requires manual bibliographic resolution before it can support a strong literature-review claim. Source rows: R0135
S9040a88e8a	unresolved_external_referenc e	Underspecification Presents Challenges for Credibility in Modern Machine Learning (2022) - D'Amour, A. et al. Underspecification Presents Challenges for Credibility in Modern Machine Learning. JMLR 23(226):1-61 (2022).	Requires manual bibliographic resolution before it can support a strong literature-review claim. Source rows: R0118;R0160
S24a406bc4c	unresolved_external_referenc e	Universal Artificial Intelligence: Sequential Decisions Based on Algorithmic Probability (2005) - Hutter, M. Universal Artificial Intelligence: Sequential Decisions Based on Algorithmic Probability. Springer (2005).	Requires manual bibliographic resolution before it can support a strong literature-review claim. Source rows: R0122
Sf7d52174e7	unresolved_external_referenc e	Variational information maximisation for intrinsically motivated reinforcement learning (2015) - Mohamed, S., & Rezende, D. J. (2015). Variational information maximisation for intrinsically motivated reinforcement learning. Advances in Neural Information Processing Systems, 28.	Requires manual bibliographic resolution before it can support a strong literature-review claim. Source rows: R0059
S9fe48dd9d0	unresolved_external_referenc e	weakly-supervised causal representation learning: interventional pairs unlock identifiability (n.d.) - Brehmer et al. weakly-supervised causal representation learning: interventional pairs unlock identifiability.	Requires manual bibliographic resolution before it can support a strong literature-review claim. Source rows: R0056
S6fe77e4d77	unresolved_external_referenc e	When Active Learning Fails, Uncalibrated Out-of-Distribution Uncertainty Epistemic uncertainty calibration: (n.d.) - When Active Learning Fails, Uncalibrated Out-of-Distribution Uncertainty Epistemic uncertainty calibration:	Requires manual bibliographic resolution before it can support a strong literature-review claim. Source rows: R0084
S23adffd29b	unresolved_external_referenc e	WILDS: A Benchmark of in-the-Wild Distribution Shifts (n.d.) - Koh, Pang Wei and Sagawa, Shiori and Marklund, Henrik and Xie, Sang Michael and Zhang, Marvin and Balsubramani, Akshay and Hu, Weihua and Yasunaga, Michihiro and Phillips, Richard Lanus and Beery, Sara and others (). WILDS: A Benchmark of in-the-Wild Distribution Shifts. International Conference on Machine Learning	Requires manual bibliographic resolution before it can support a strong literature-review claim. Source rows: R0165

Appendix C: Abstract-Resolved External Sources

Title	Topics	Evidence note
Designing and Interpreting Probes with Control Tasks (2019)	agency_viability	John Hewitt, Percy Liang. Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP).
Do As I Can, Not As I Say: Grounding Language in Robotic Affordances (2022)	agency_viability	We propose to provide real-world grounding by means of pretrained skills, which are used to constrain the model to propose natural language actions that are both feasible and contextually appropriate. We show how low-level skills can be combined with large language models so that the language model provides high-level knowledge about the procedures for performing complex and temporally-extended instructions, while value functions associated with these skills provide the grounding necessary to connect this...
AgentBench: Evaluating LLMs as Agents (2023)	agency_viability	Our extensive test over \num API-based and open-sourced (OSS) LLMs shows that, while top commercial LLMs present a strong ability of acting as agents in complex environments, there is a significant disparity in performance between them and many OSS competitors that are no larger than 70B. We identify the typical reasons of failures in environments and LLMs, showing that poor long-term reasoning, decision-making, and instruction following abilities are the main obstacles for developing usable LLM agents.
Technological Approach to Mind Everywhere: An Experimentally-Grounded Framework for Understanding Diverse Bodies and Minds (2021)	agency_viability, geometry_symmetry, meaning_mind	In this Perspective, I introduce TAME-Technological Approach to Mind Everywhere-a framework for understanding and manipulating cognition in unconventional substrates.
Goal Misgeneralization in Deep Reinforcement Learning (2021)	agency_viability, ood_generalization	We study goal misgeneralization, a type of out-of-distribution generalization failure in reinforcement learning (RL). Goal misgeneralization failures occur when an RL agent retains its capabilities out-of-distribution yet pursues the wrong goal.
WebArena: A Realistic Web Environment for Building Autonomous Agents (2023)	agency_viability, ood_generalization, world_models	The results demonstrate that solving complex tasks is challenging: our best GPT-4-based agent only achieves an end-to-end task success rate of 14.41%, significantly lower than the human performance of 78.24%.
Empowerment: A Universal Agent-Centric Measure of Control (2005)	agency_viability, uncertainty_inquiry	Inspired by examples from the animal kingdom, social sciences and games we propose empowerment, a rather universal function, defined as the information-theoretic capacity of an agent's actuation channel. Using two simple experiments we also demonstrate how empowerment influences sensor-actuator evolution
Empowerment: A Universal Agent-Centric Measure of Control (2005)	agency_viability, uncertainty_inquiry	Inspired by examples from the animal kingdom, social sciences and games we propose empowerment, a rather universal function, defined as the information-theoretic capacity of an agent's actuation channel. Using two simple experiments we also demonstrate how empowerment influences sensor-actuator evolution
Reflexion: Language Agents with Verbal Reinforcement Learning (2023)	agency_viability, uncertainty_inquiry	We propose Reflexion, a novel framework to reinforce language agents not by updating weights, but instead through linguistic feedback.
Curiosity-Driven Exploration by Self-Supervised Prediction (2017)	agency_viability, uncertainty_inquiry, ood_generalization	The proposed approach is evaluated in two environments: VizDoom and Super Mario Bros.
World Models (2018)	agency_viability, uncertainty_inquiry, world_models	We explore building generative neural network models of popular reinforcement learning environments. Our world model can be trained quickly in an unsupervised manner to learn a compressed spatial and temporal representation of the environment.
CausaLab: A Scalable Environment for Interactive Causal Discovery Toward AI Scientists (2026)	causal_representation, agency_viability	We introduce CausaLab, a scalable environment for evaluating interactive causal discovery by LLM agents. Experiments show a persistent gap between prediction and mechanism recovery: in the purely observational 6-node setting, GPT-5.2-high reaches 92% task accuracy but only 0.471 all-edge \$F_{1}\$.
Autopoiesis and Cognition: The Realization of the Living (1980)	causal_representation, agency_viability, meaning_mind, world_models	Editorial Preface General Table Of Contents Foreword Introduction (by Professor Maturana) Biology Of Cognition Dedication Table of Contents I. Introduction II.
Object-Centric World Models for Causality-Aware Reinforcement Learning (2025)	causal_representation, agency_viability, meaning_mind, world_models	Motivated by this insight, we propose \emph{Slot Transformer Imagination with CAusality-aware reinforcement learning} (STICA), a unified framework in which object-centric Transformers serve as the world model and causality-aware policy and value networks. Experiments on object-rich benchmarks demonstrate that STICA consistently outperforms state-of-the-art agents in both sample efficiency and final performance.
Voyager: An Open-Ended Embodied Agent with Large Language Models (2023)	causal_representation, agency_viability, uncertainty_inquiry	We introduce Voyager, the first LLM-powered embodied lifelong learning agent in Minecraft that continuously explores the world, acquires diverse skills, and makes novel discoveries without human intervention. Empirically, Voyager shows strong in-context lifelong learning capability and exhibits exceptional proficiency in playing Minecraft.
Latent State Design for World Models under Sufficiency Constraints (2026)	causal_representation, agency_viability, uncertainty_inquiry, geometry_symmetry, ood_generalization, world_models	We propose a functional taxonomy that groups methods by what their latent state is for, rather than by architecture or application domain: predictive embedding, recurrent belief state, object/causal structure, latent action interface, grounded planning interface, and memory substrate.
Why LLMs Fail at Causal Discovery and How Interventional Agents Escape (2026)	causal_representation, agency_viability, uncertainty_inquiry, geometry_symmetry, world_models	Recent benchmarks show that even fine-tuned models plateau on simple causal graphs and degrade as complexity grows, but why they fail has not been established. We propose Agentic Causal Bayesian Optimization (A-CBO), wherein a frozen language model serves as an interventional oracle answering targeted queries about intervention effects, while an external Bayesian loop concentrates beliefs over candidate graphs in logarithmically many rounds.

Title	Topics	Evidence note
OSWorld: Benchmarking Multimodal Agents for Open-Ended Tasks in Real Computer Environments (2024)	causal_representation, agency_viability, uncertainty_inquiry, ood_generalization	To address this issue, we introduce OSWorld, the first-of-its-kind scalable, real computer environment for multimodal agents, supporting task setup, execution-based evaluation, and interactive learning across various operating systems such as Ubuntu, Windows, and macOS.
Active Causal Experimentalist (ACE): Learning Intervention Strategies via Direct Preference Optimization (2026)	causal_representation, agency_viability, uncertainty_inquiry, ood_generalization, meaning_mind	We propose Active Causal Experimentalist (ACE), which learns experimental design as a sequential policy.
Active Inference and Epistemic Value in Graphical Models (2022)	causal_representation, agency_viability, uncertainty_inquiry, world_models	The Free Energy Principle (FEP) postulates that biological agents perceive and interact with their environment in order to minimize a Variational Free Energy (VFE) with respect to a generative model of their environment. The inference of a policy (future control sequence) according to the FEP is known as Active Inference (AIF).
Comments on the paper Causal inference by using invariant prediction: identification and confidence intervals by Peters, Buhlmann and Meinshausen (2016)	causal_representation, geometry_symmetry, ood_generalization	Here, we propose to exploit this invariance of a prediction under a causal model for causal inference: given different experimental settings (e.g. various interventions) we collect all models that do show invariance in their predictive accuracy across settings and interventions.
On the Ability of Deep Networks to Learn Symmetries from Data: A Neural Kernel Theory (2024)	causal_representation, geometry_symmetry, ood_generalization	The group-cyclic nature of the dataset allows us to analyze the Gram matrix of neural kernels in the Fourier domain; here we find a simple characterization of the generalization error as a function of class separation (signal) and class-orbit density (noise).
Invariant Risk Minimization (2019)	causal_representation, geometry_symmetry, ood_generalization, world_models	We introduce Invariant Risk Minimization (IRM), a learning paradigm to estimate invariant correlations across multiple training distributions. Through theory and experiments, we show how the invariances learned by IRM relate to the causal structures governing the data and enable out-of-distribution generalization.
Toward Causal Representation Learning (2021)	causal_representation, ood_generalization, world_models	Finally, we delineate some implications of causality for machine learning and propose key research areas at the intersection of both communities.
Inducing Causal World Models in LLMs for Zero-Shot Physical Reasoning (2025)	causal_representation, ood_generalization, world_models	In this paper, we introduce Causal World Model Induction (CWMI), a novel framework designed to embed an explicit model of causal physics within an LLM. Experimental results show that CWMI significantly outperforms state-of-the-art LLMs on zero-shot physical reasoning tasks, including the PIQA benchmark and our newly proposed PhysiCa-Bench dataset.
Compositional Neuro-Symbolic Reasoning (2026)	causal_representation, ood_generalization, world_models	We therefore propose a neuro-symbolic architecture that extracts object-level structure from grids, uses neural priors to propose candidate transformations from a fixed domain-specific language (DSL) of atomic patterns, and filters hypotheses using cross-example consistency. These results demonstrate that separating perception, neural-guided transformation proposal, and symbolic consistency filtering improves generalization without task-specific finetuning or reinforcement learning, while reducing reliance on...
Locating and Editing Factual Associations in GPT (2022)	causal_representation, uncertainty_inquiry, ood_generalization	We analyze the storage and recall of factual associations in autoregressive transformer language models, finding evidence that these associations correspond to localized, directly-editable computations. We find that ROME is effective on a standard zero-shot relation extraction (zsRE) model-editing task, comparable to existing methods.
Weakly supervised causal representation learning (2022)	causal_representation, uncertainty_inquiry, world_models	We then introduce implicit latent causal models, variational autoencoders that represent causal variables and causal structure without having to optimize an explicit discrete graph structure. On simple image data, including a novel dataset of simulated robotic manipulation, we demonstrate that such models can reliably identify the causal structure and disentangle causal variables.
Chapman and Hall (1985)	general_foundation	Journal Article Chapman and Hall Get access Anne Lohrli Anne Lohrli Claremont, California Search for other works by this author on: Oxford Academic Google Scholar Notes and Queries, Volume 32, Issue 3, September 1985, Pages 377-378, https://doi.org/10.1093/notesj/32.3.377-a Published: 01 September 1985
Abandoning Objectives: Evolution Through the Search for Novelty Alone (2010)	general_foundation	This paper proposes an approach to circumventing deception that also yields a new perspective on open-ended evolution.
Neural networks are a priori biased towards Boolean functions with low entropy (2019)	general_foundation	Finally, we prove that, regardless of the distribution of inputs, the bias towards low entropy becomes monotonically stronger upon adding ReLU layers, and empirically show that increasing the variance of the bias term has a similar effect.
Beyond the Imitation Game: Quantifying and extrapolating the capabilities of language models (2022)	general_foundation	Language models demonstrate both quantitative improvement and new qualitative capabilities with increasing scale. To address this challenge, we introduce the Beyond the Imitation Game benchmark (BIG-bench).
Learning Layer-wise Equivariances Automatically using Gradients (2023)	geometry_symmetry	We demonstrate the ability to automatically learn layer-wise equivariances on image classification tasks, achieving equivalent or improved performance over baselines with hard-coded symmetry.
Are Flat Minima an Illusion? (2026)	geometry_symmetry, description_length	Here I show that the actual driver is weakness, the volume of completions compatible with the learned function in the learner's embodied language.
Group Equivariant Convolutional Networks (2016)	geometry_symmetry, ood_generalization	We introduce Group equivariant Convolutional Neural Networks (G-CNNs), a natural generalization of convolutional neural networks that reduces sample complexity by exploiting symmetries.
Neural Language of Thought Models (2024)	geometry_symmetry, ood_generalization, meaning_mind, world_models	In this work, we introduce the Neural Language of Thought Model (NLotM), a novel approach for unsupervised learning of LoTH-inspired representation and generation.

Title	Topics	Evidence note
Distributionally Robust Neural Networks for Group Shifts: On the Importance of Regularization for Worst-Case Generalization (2019)	geometry_symmetry, ood_generalization, world_models	However, we find that naively applying group DRO to overparameterized neural networks fails: these models can perfectly fit the training data, and any model with vanishing average training loss also already has vanishing worst-case training loss. Finally, we introduce a stochastic optimization algorithm, with convergence guarantees, to efficiently train group DRO models.
Representation Learning of Geometric Trees (2024)	geometry_symmetry, ood_generalization, world_models	To address this, we introduce a new representation learning framework tailored for geometric trees.
Geometric Deep Learning: Grids, Groups, Graphs, Geodesics, and Gauges (2021)	geometry_symmetry, world_models	The last decade has witnessed an experimental revolution in data science and machine learning, epitomised by deep learning methods. Indeed, many high-dimensional learning tasks previously thought to be beyond reach -- such as computer vision, playing Go, or protein folding -- are in fact feasible with appropriate computational scale.
Design for a Brain (1952)	meaning_mind	New findings concerning brain and behavior have forced upon us unpleasant necessity of reexamining our models. The most striking is that manifested by living organism, where we find that from a random collection of atoms there is a synthesis of cells which in turn arrange themselves in most intricate orderly system of an organism.
On the Computation of Meaning, Language Models and Incomprehensible Horrors (2023)	meaning_mind, world_models	To address this, we propose simulating human feelings and optimising models to construct weak representations. Our findings shed light on the relationship between meaning and intelligence, and how we can build machines that comprehend and intend meaning.
Illuminating search spaces by mapping elites (2015)	ood_generalization	Many fields use search algorithms, which automatically explore a search space to find high-performing solutions: chemists search through the space of molecules to discover new drugs; engineers search for stronger, cheaper, safer designs, scientists search for models that best explain data, etc. Interestingly, because MAP-Elites explores more of the search space, it also tends to find a better overall solution than state-of-the-art search algorithms.
Shortcut Learning in Deep Neural Networks (2020)	ood_generalization	Deep learning has triggered the current rise of artificial intelligence and is the workhorse of today's machine intelligence. Numerous success stories have rapidly spread all over science, industry and society, but its limitations have only recently come into focus.
Grokking: Generalization Beyond Overfitting on Small Algorithmic Datasets (2022)	ood_generalization	In this paper we propose to study generalization of neural networks on small algorithmically generated datasets. In some situations we show that neural networks learn through a process of "grokking" a pattern in the data, improving generalization performance from random chance level to perfect generalization, and that this improvement in generalization can happen well past the point of overfitting.
Domain Generalization: A Survey (2021)	ood_generalization, meaning_mind	Generalization to out-of-distribution (OOD) data is a capability natural to humans yet challenging for machines to reproduce. This is because most learning algorithms strongly rely on the i.i.d.-assumption on source/target data, which is often violated in practice due to domain shift.
Diff-Plugin: Revitalizing Details for Diffusion-based Low-level Tasks (2024)	ood_generalization, meaning_mind	Specifically, we first propose a lightweight Task-Plugin module with a dual branch design to provide task-specific priors, guiding the diffusion process in preserving image content. We then propose a Plugin-Selector that can automatically select different Task-Plugins based on the text instruction, allowing users to edit images by indicating multiple low-level tasks with natural language.
In Search of Lost Domain Generalization (2020)	ood_generalization, world_models	Contrary to prior work, we argue that domain generalization algorithms without a model selection strategy should be regarded as incomplete. We conduct extensive experiments using DomainBed and find that, when carefully implemented, empirical risk minimization shows state-of-the-art performance across all datasets.
GAIA: a benchmark for General AI Assistants (2023)	ood_generalization, world_models	We introduce GAIA, a benchmark for General AI Assistants that, if solved, would represent a milestone in AI research. GAIA proposes real-world questions that require a set of fundamental abilities such as reasoning, multi-modality handling, web browsing, and generally tool-use proficiency.
Active Learning Literature Survey (2009)	uncertainty_inquiry	This includes a discussion of the scenarios in which queries can be formulated, and an overview of the query strategy frameworks proposed in the literature to date.
Bayesian Active Learning for Classification and Preference Learning (2011)	uncertainty_inquiry	We propose an approach that expresses information gain in terms of predictive entropies, and apply this method to the Gaussian Process Classifier (GPC).
Le normal et le pathologique (2013)	uncertainty_inquiry	Le travail prsent est un effort pour intgrer la spculation philosophique quelques-unes des mthodes et des acquisitions de la mdecine. Il ne s'agit de donner aucune leon, de porter sur l'activit mdicale aucun jugement normatif [].
Deep Bayesian Active Learning with Image Data (2017)	uncertainty_inquiry	Taking advantage of specialised models such as Bayesian convolutional neural networks, we demonstrate our active learning techniques with image data, obtaining a significant improvement on existing active learning approaches. We demonstrate this on both the MNIST dataset, as well as for skin cancer diagnosis from lesion images (ISIC2016 task).
Toolformer: Language Models Can Teach Themselves to Use Tools (2023)	uncertainty_inquiry	In this paper, we show that LMs can teach themselves to use external tools via simple APIs and achieve the best of both worlds. We introduce Toolformer, a model trained to decide which APIs to call, when to call them, what arguments to pass, and how to best incorporate the results into future token prediction.
Structuring Open-Ended NAS: Semi-Automated Design Knowledge Structuring with LLMs for Efficient Neural Architecture Search (2026)	uncertainty_inquiry	To address these issues, we propose to semi-automatically structure model design knowledge to guide the search process. To efficiently explore this vast space, we introduce FairNAD, using a multi-type mutation that enables broad exploration through mutation with fair idea sampling, Pareto-aware mutation, LLM-driven iterative mutation, and a fine-grained feedback loop.

Title	Topics	Evidence note
Bayesian Active Learning by Disagreements: A Geometric Perspective (2021)	uncertainty_inquiry, geometry_symmetry	Experiments show that GBALD has slight perturbations to noisy and repeated samples, and outperforms BALD, BatchBALD and other existing deep active learning approaches.
Dropout as a Bayesian Approximation: Representing Model Uncertainty in Deep Learning (2015)	uncertainty_inquiry, meaning_mind	We show a considerable improvement in predictive log-likelihood and RMSE compared to existing state-of-the-art methods, and finish by using dropout's uncertainty in deep reinforcement learning.
Concrete Problems in AI Safety (2016)	uncertainty_inquiry, meaning_mind	Rapid progress in machine learning and artificial intelligence (AI) has brought increasing attention to the potential impacts of AI technologies on society. In this paper we discuss one such potential impact: the problem of accidents in machine learning systems, defined as unintended and harmful behavior that may emerge from poor design of real-world AI systems.
Simple and Scalable Predictive Uncertainty Estimation using Deep Ensembles (2016)	uncertainty_inquiry, ood_generalization	We propose an alternative to Bayesian NNs that is simple to implement, readily parallelizable, requires very little hyperparameter tuning, and yields high quality predictive uncertainty estimates. Through a series of experiments on classification and regression benchmarks, we demonstrate that our method produces well-calibrated uncertainty estimates which are as good or better than approximate Bayesian NNs.
What Uncertainties Do We Need in Bayesian Deep Learning for Computer Vision? (2017)	uncertainty_inquiry, ood_generalization	There are two major types of uncertainty one can model. Aleatoric uncertainty captures noise inherent in the observations.
Holistic Evaluation of Language Models (2022)	uncertainty_inquiry, ood_generalization	Our evaluation surfaces 25 top-level findings.
Deep learning generalizes because the parameter-function map is biased towards simple functions (2018)	uncertainty_inquiry, ood_generalization, description_length	By applying a very general probability-complexity bound recently derived from algorithmic information theory (AIT), we argue that the parameter-function map of many DNNs should be exponentially biased towards simple functions.
Decomposed Prompting: A Modular Approach for Solving Complex Tasks (2022)	uncertainty_inquiry, ood_generalization, world_models	To address this, we propose Decomposed Prompting, a new approach to solve complex tasks by decomposing them (via prompting) into simpler sub-tasks that can be delegated to a library of prompting-based LLMs dedicated to these sub-tasks. We show that the flexibility and modularity of Decomposed Prompting allows it to outperform prior work on few-shot prompting using GPT3.
Exploration by Random Network Distillation (2018)	uncertainty_inquiry, world_models	We introduce an exploration bonus for deep reinforcement learning methods that is easy to implement and adds minimal overhead to the computation performed. We also introduce a method to flexibly combine intrinsic and extrinsic rewards.
ReAct: Synergizing Reasoning and Acting in Language Models (2022)	uncertainty_inquiry, world_models	While large language models (LLMs) have demonstrated impressive capabilities across tasks in language understanding and interactive decision making, their abilities for reasoning (e.g. We apply our approach, named ReAct, to a diverse set of language and decision making tasks and demonstrate its effectiveness over state-of-the-art baselines, as well as improved human interpretability and trustworthiness over methods without reasoning or acting components.
Tree of Thoughts: Deliberate Problem Solving with Large Language Models (2023)	uncertainty_inquiry, world_models	To surmount these challenges, we introduce a new framework for language model inference, Tree of Thoughts (ToT), which generalizes over the popular Chain of Thought approach to prompting language models, and enables exploration over coherent units of text (thoughts) that serve as intermediate steps toward problem solving. Our experiments show that ToT significantly enhances language models' problem-solving abilities on three novel tasks requiring non-trivial planning or search: Game of 24, Creative Writing, and...
Challenging Common Assumptions in the Unsupervised Learning of Disentangled Representations (2018)	world_models	We first theoretically show that the unsupervised learning of disentangled representations is fundamentally impossible without inductive biases on both the models and the data.
SWE-bench: Can Language Models Resolve Real-World GitHub Issues? (2023)	world_models	We find real-world software engineering to be a rich, sustainable, and challenging testbed for evaluating the next generation of language models. To this end, we introduce SWE-bench, an evaluation framework consisting of \$2,294\$ software engineering problems drawn from real GitHub issues and corresponding pull requests across \$12\$ popular Python repositories.